

1 Which quantity is a vector?

- A pressure
- B temperature
- C weight
- D work

2 Four students, **A**, **B**, **C** and **D**, have completed an experiment to determine the acceleration of free fall, g . Each student repeated the experiment three times. The determined values of g are shown in the table.

Which set of results has a high precision and a low accuracy?

	acceleration of free fall / m s^{-2}		
	experiment 1	experiment 2	experiment 3
A	7.2	9.4	8.3
B	9.5	9.8	10.2
C	9.8	9.8	9.9
D	10.1	10.2	10.1

3 The diameter of a ball is measured as $(5.26 \pm 0.02) \text{ cm}$.

What is the absolute uncertainty in the volume of the ball?

- A 0.29 cm^3
- B 0.87 cm^3
- C 1.1 cm^3
- D 7.0 cm^3